US007913963B2

(12) **United States Patent**
Cheng et al.

(10) **Patent No.: US 7,913,963 B2**
(45) **Date of Patent: Mar. 29, 2011**

(54) **DEVICE STAND WITH SUCTION BASE**

(75) Inventors: **Gang Cheng**, Shenzhen (CN); **Meng-Hua He**, Shenzhen (CN); **Chun-Che Yen**, Taipei Hsien (TW); **Yu-Tao Chen**, Taipei Hsien (TW); **Te-Sheng Jan**, Taipei Hsien (TW)

(73) Assignees: **Hong Fu Jin Precision Industry (ShenZhen) Co., Ltd.**, Shenzhen, Guangdong Province (CN); **Hon Hai Precision Industry Co., Ltd.**, Tu-Cheng, New Taipei (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 174 days.

(21) Appl. No.: **12/427,705**

(22) Filed: **Apr. 21, 2009**

(65) **Prior Publication Data**

US 2010/0044535 A1 Feb. 25, 2010

(30) **Foreign Application Priority Data**

Aug. 22, 2008 (CN) 2008 1 0304154

(51) **Int. Cl.**
A45D 42/14 (2006.01)

(52) **U.S. Cl.** **248/205.5**; 248/206.2; 248/363

(58) **Field of Classification Search** 248/176.1, 248/205.2, 205.6, 205.7, 205.8, 206.2, 309.1, 248/309.3, 311.2, 363
See application file for complete search history.

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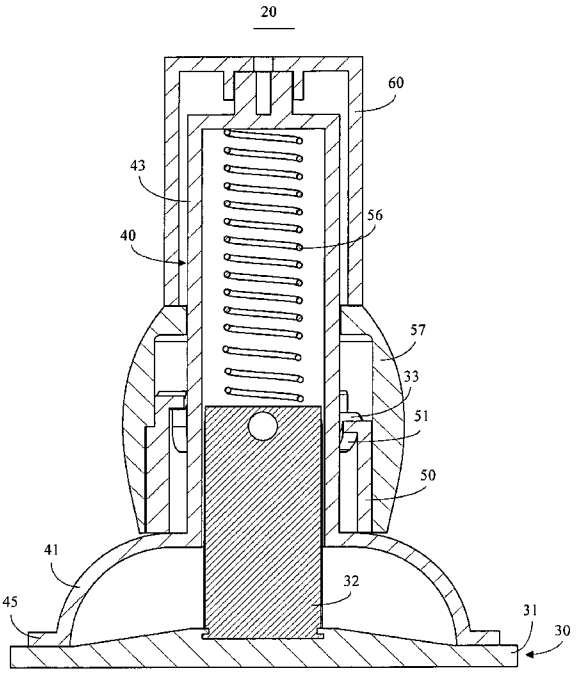
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Primary Examiner — Gwendolyn Baxter
(74) *Attorney, Agent, or Firm* — Zhigang Ma

(57) **ABSTRACT**

A device stand includes a suction base and a load-supporting module mounted on the suction base. The suction base includes a suction piece, a fixing structure, a face cam, pins, and a fixing cap. The fixing structure, configured for fitting over the suction piece, includes a shaft housing and a sleeve extending upwardly from the shaft housing. The sleeve defines axial running slots. The face cam fit over the sleeve of the fixing structure. One end of each of the pins engages with the face of the face cam, the other end of each of the pins passes through the axial running slots of the sleeve and connects with the shaft of the suction piece. The fixing cap is mounted on an upper end of the sleeve. The stand can be fixed to/taken off from a flat surface easily.

10 Claims, 7 Drawing Sheets



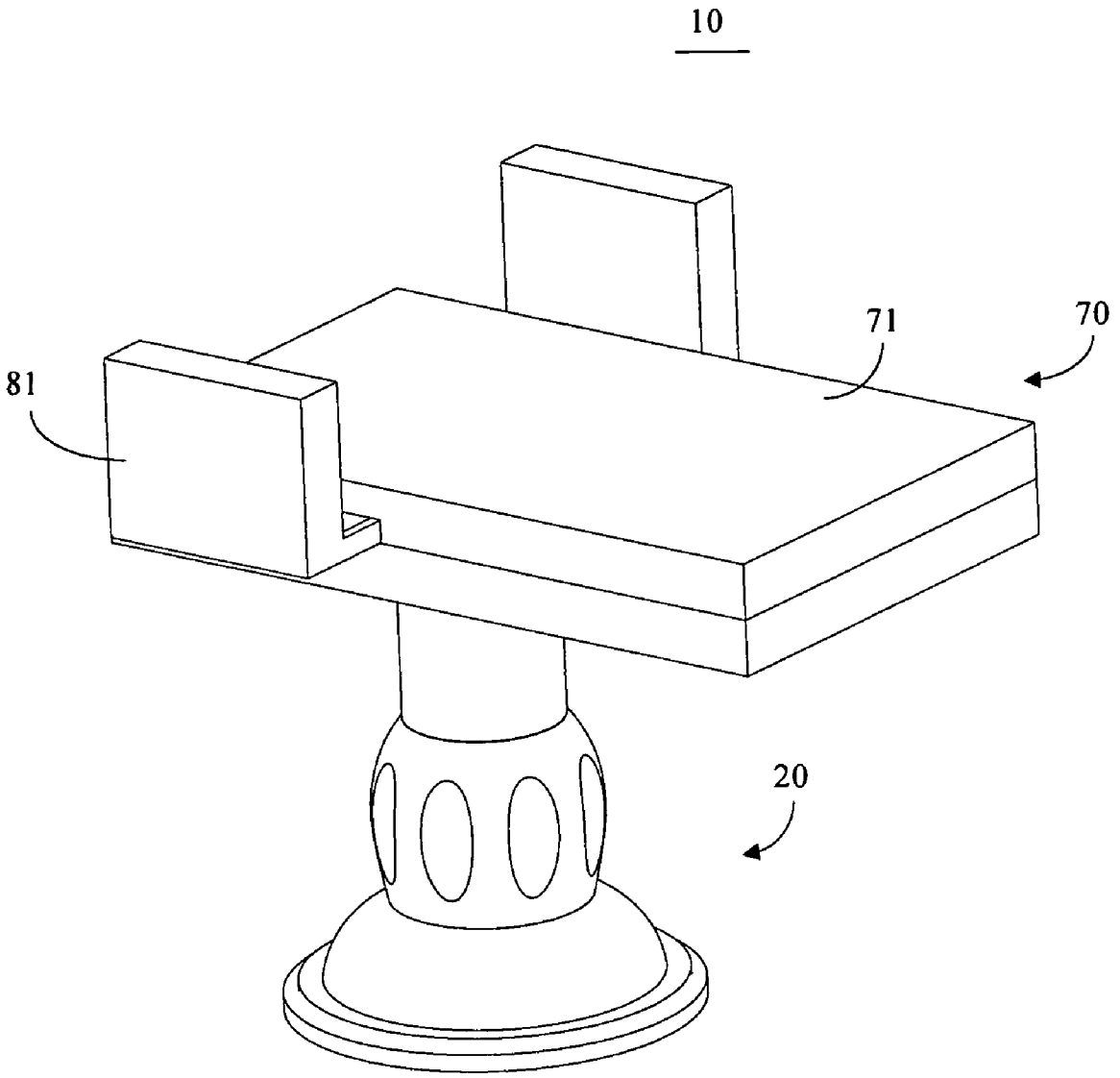


FIG. 1

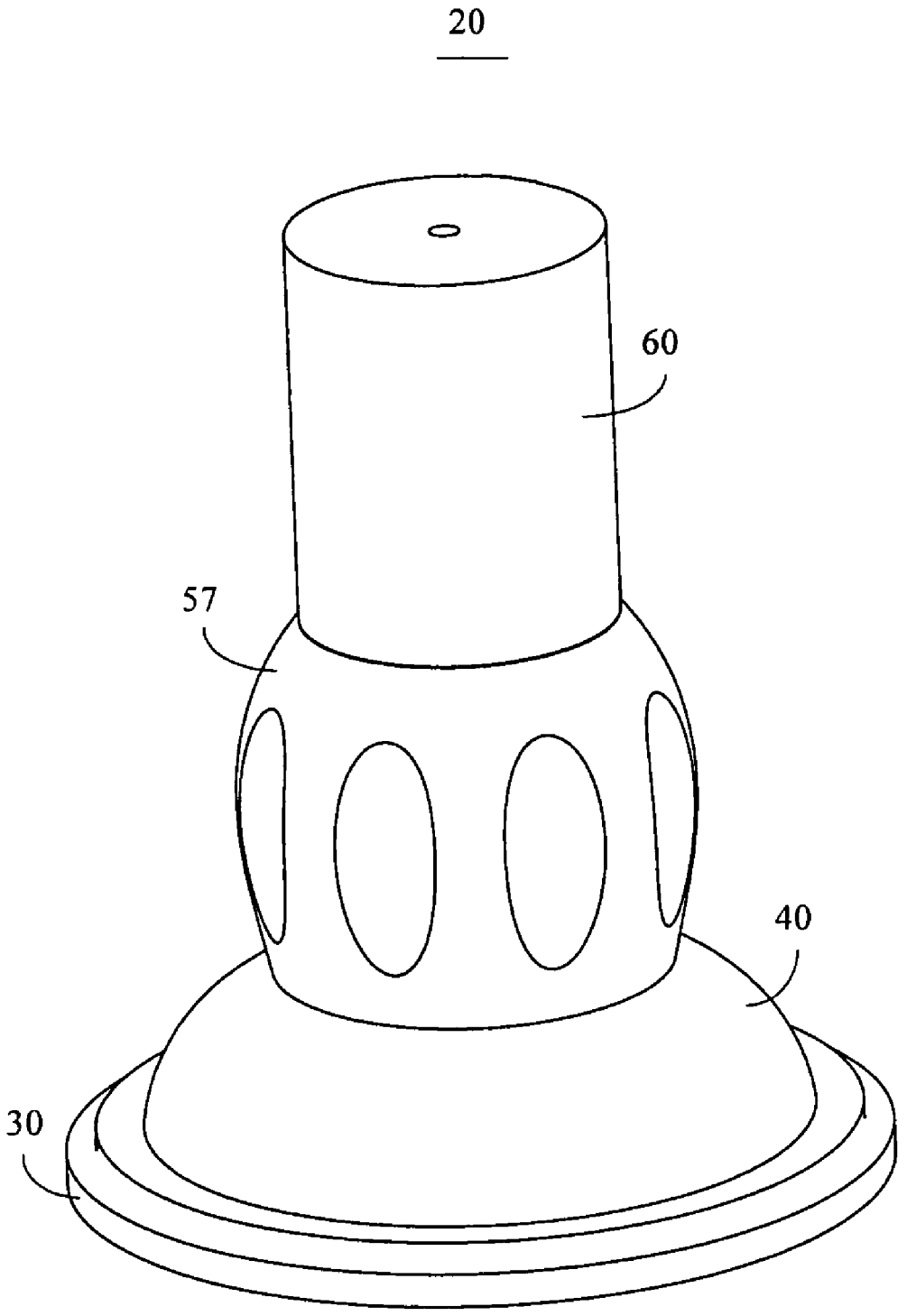


FIG. 2

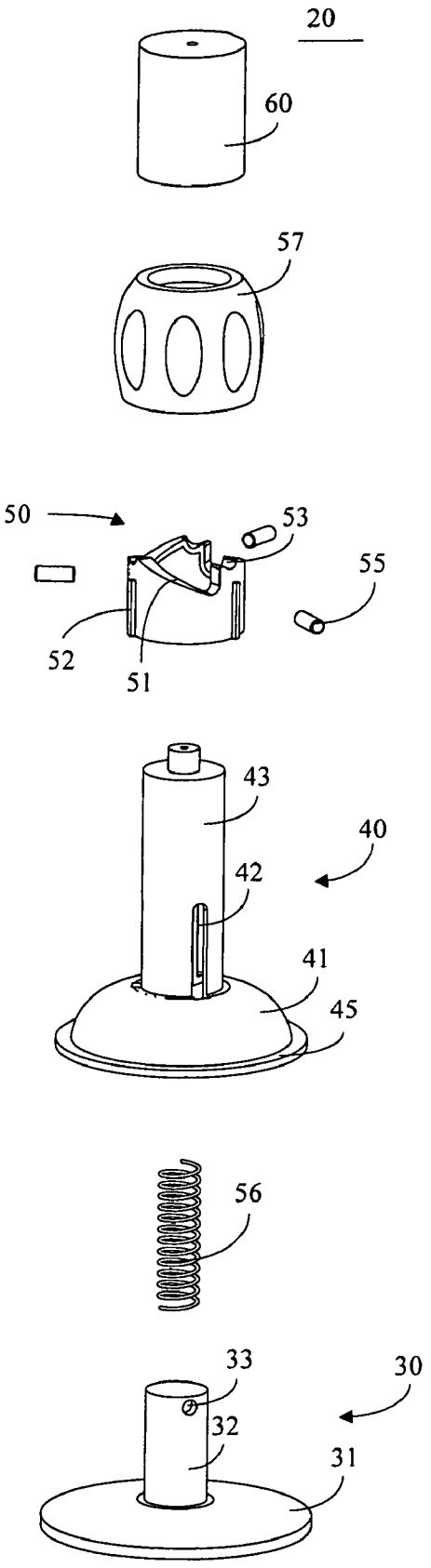


FIG. 3

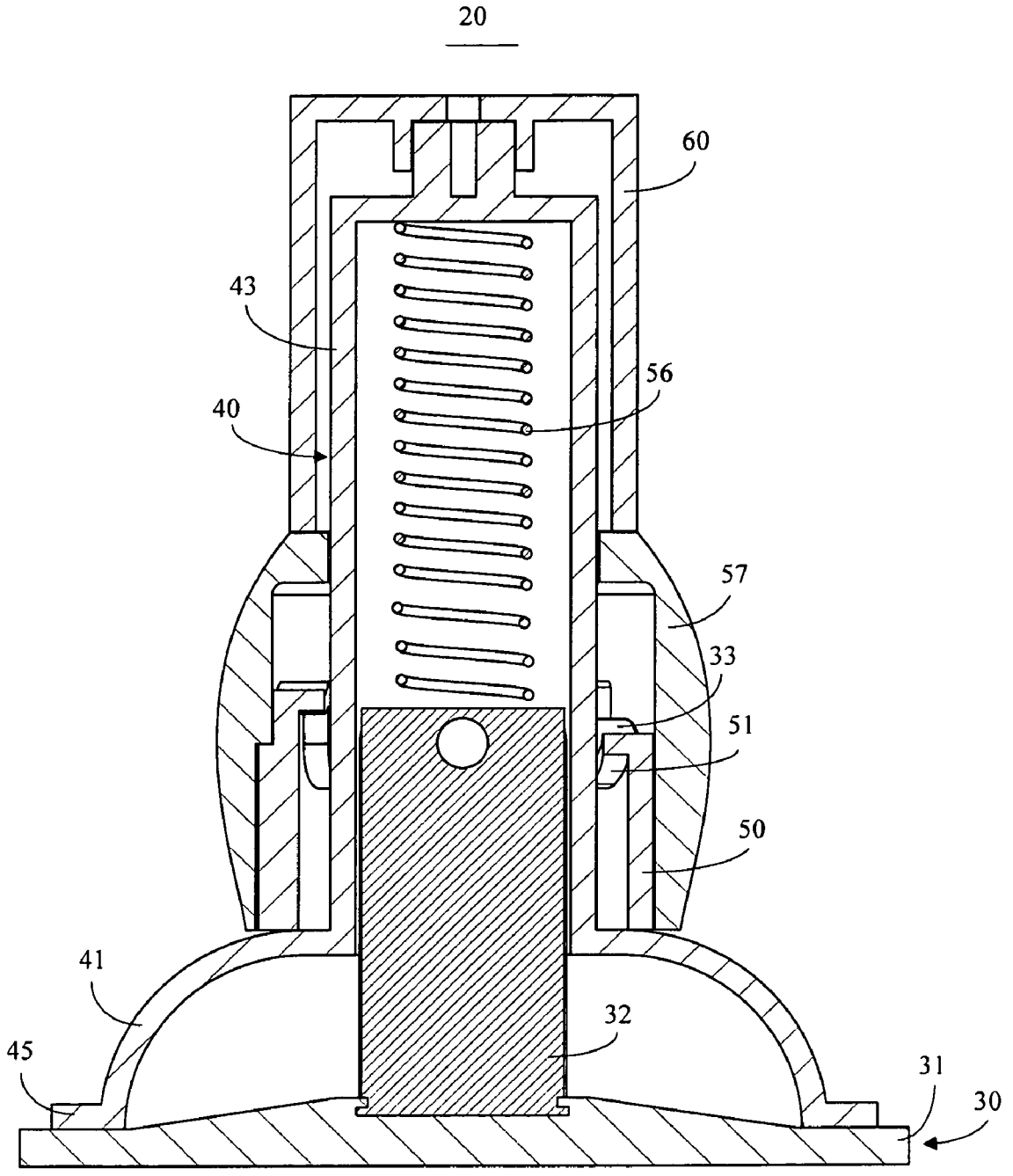


FIG. 4

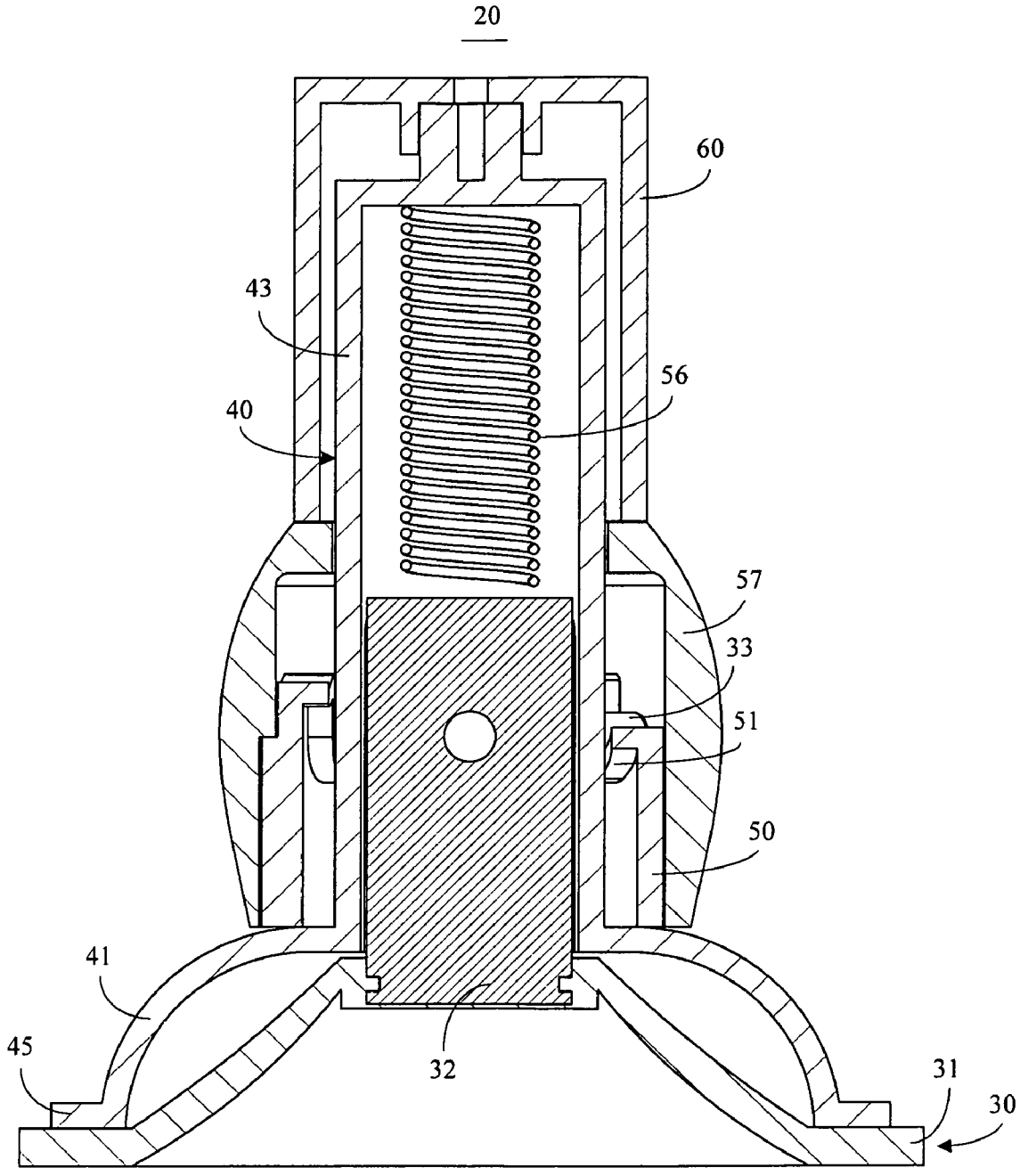


FIG. 5

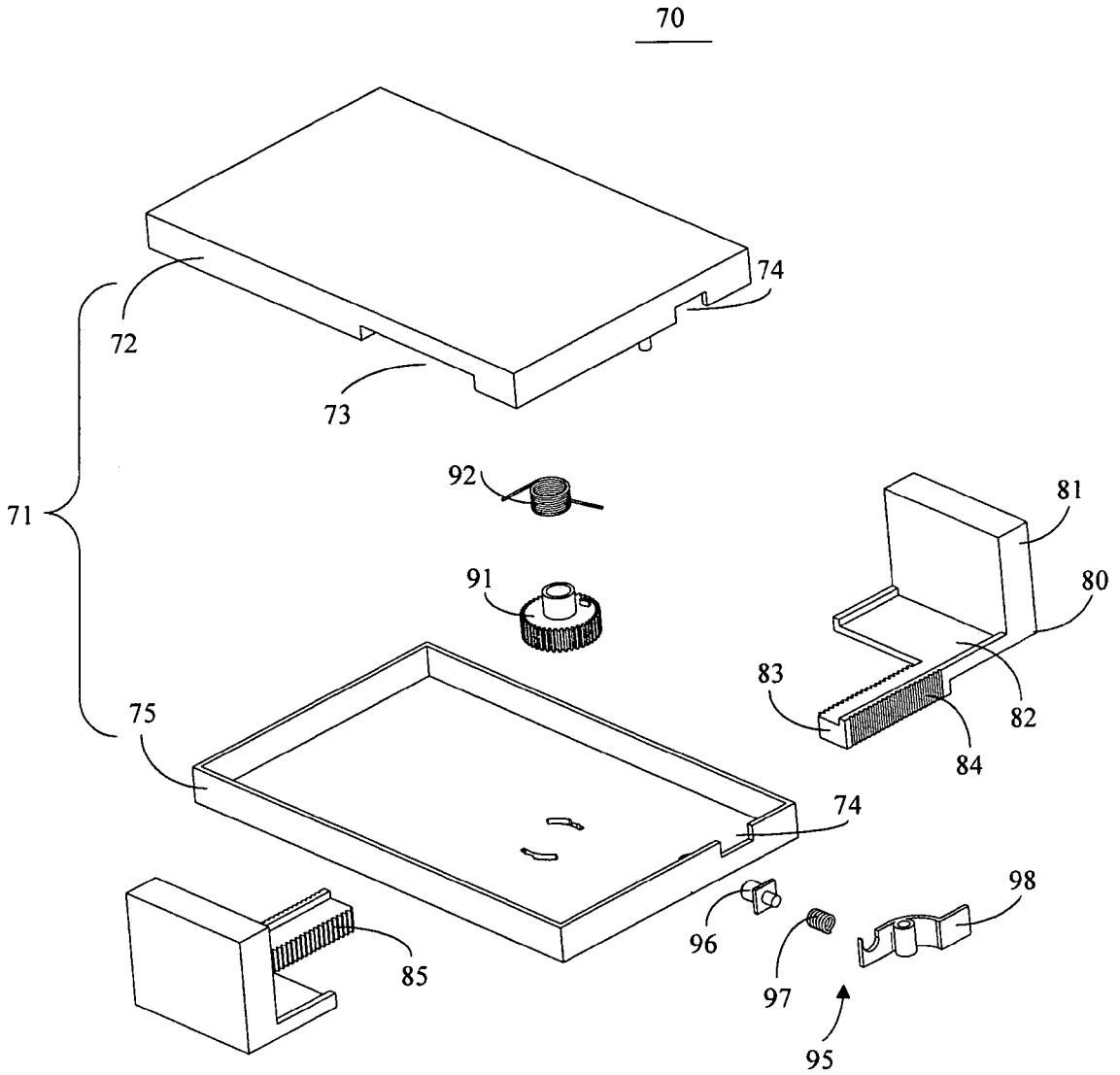


FIG. 6

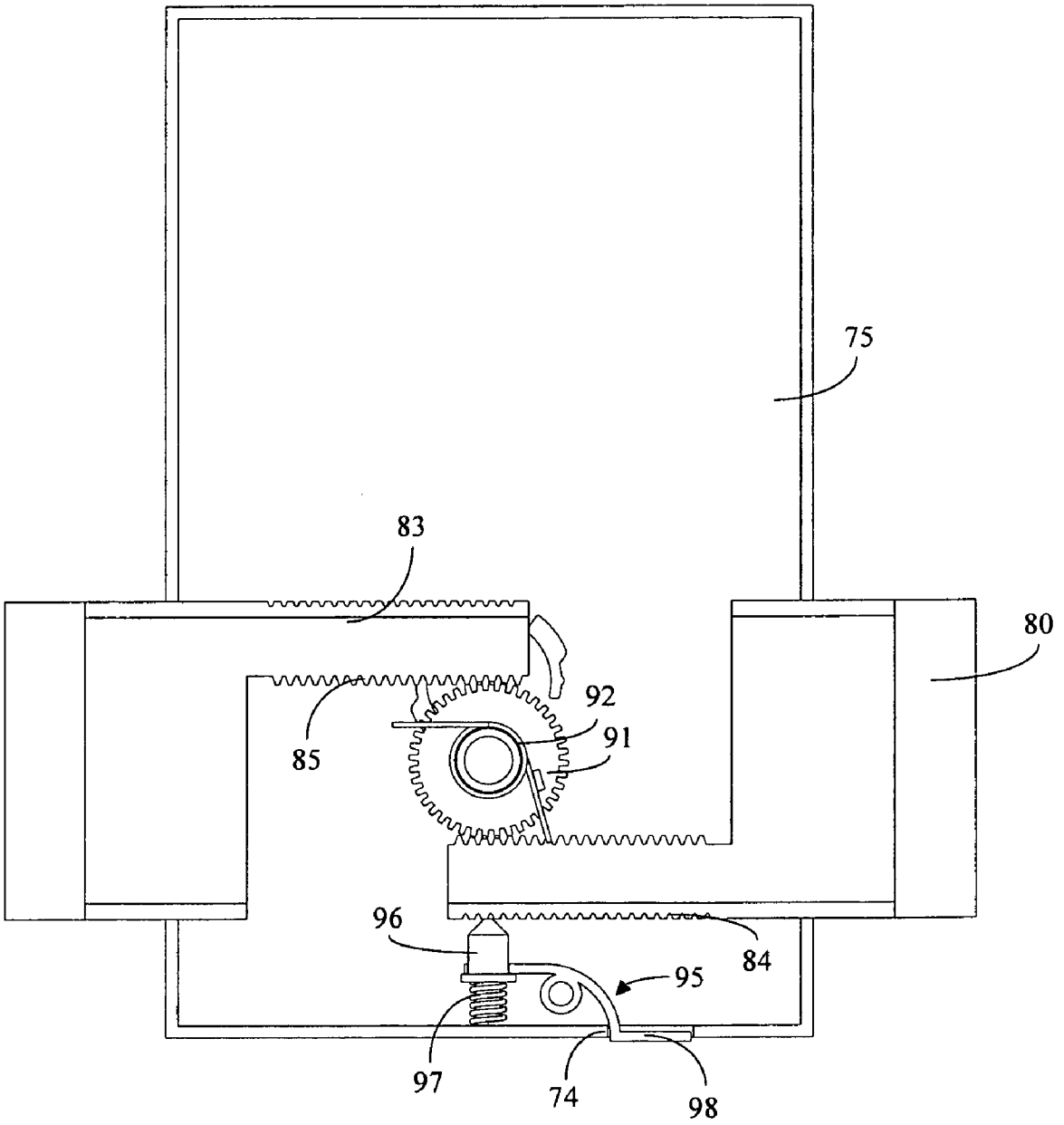


FIG. 7

DEVICE STAND WITH SUCTION BASE

BACKGROUND

1. Technical Field

The present disclosure relates to a device stand with a suction base.

2. Description of Related Art

A motor vehicle may include a stand for supporting a mini computer, a mobile navigation apparatus, or a mobile phone. The stand is usually fixed on a surface inside the motor vehicle with fasteners or adhesive and cannot be easily moved if necessary.

What is needed is a stand that can be easily relocated.

BRIEF DESCRIPTION OF THE DRAWINGS

Many aspects of the embodiments can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the present disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

FIG. **1** is a schematic, isometric view of a stand according to an exemplary embodiment, the stand including a suction base and a load-supporting module.

FIG. **2** is a schematic, isometric view of the suction base of FIG. **1**.

FIG. **3** is an exploded view of the suction base of FIG. **2**.

FIG. **4** is a section view of the suction base of FIG. **2** without a vacuum seal created.

FIG. **5** is a section view of the suction base of FIG. **2** with a vacuum seal created.

FIG. **6** is a exploded view of the load-supporting module of FIG. **1**.

FIG. **7** is a top plan view of the load-supporting module of FIG. **1**, with a top cover removed.

DETAILED DESCRIPTION

Referring to FIG. **1**, a stand **10** according to an exemplary embodiment is disclosed. The stand **10** includes a suction base **20** and a load-supporting module **70** mounted on top of the suction base **20**.

Referring to FIGS. **2**, **3** and **4**, the suction base **20** includes a suction piece **30**, a spring **56**, a fixing structure **40** for fitting over the suction piece **30**, pins **55**, a face cam **50**, a rotatable sleeve **57**, and a fixing cap **60**.

The suction piece **30** includes a membrane **31** and a shaft **32** extending upwardly from the membrane **31**.

The fixing structure **40** includes a shaft housing **41** and a sleeve **43** extending upwardly from the shaft housing **41**, and lower edge of the housing **41** forms a flange **45**. The sleeve **43** defines axial running slots **42**. An upper end of the sleeve **43** is closed.

The face cam **50** is ring-shaped and has a face **51** which includes grooves **53** defined therein. The face cam **50** further includes axial rails **52** projected from an outer circumferential surface thereof.

During assembly of the suction base **20**, first, the sleeve **43** receives the spring **56**, then is fit over the shaft **32**. The spring **56** is positioned between the upper end of the sleeve **43** and the shaft **32**. Second, the face cam **50** is fit over the sleeve **43**. One end of each of the pins **55** engages with the face **51** of the face cam **55**. The other end of each of the pins **55** extends through the running slots **42** defined in the sleeve **43** and enters into holes **33** defined in the shaft **32**. Third, the rotatable sleeve **57** fits over the face cam **50**, and grooves (not shown) defined in an inner surface of the rotatable sleeve **57** engage with the rails **52** of the face cam **50**, thus to connect the rotatable sleeve **57** to the face cam **50**. Finally, the fixing cap **60** is fixed to the upper end of the sleeve **43** by screws (not shown).

Referring to FIGS. **4** and **5**, in use, the suction base **20** is put on a flat surface, such as a window, ensuring the membrane **31** is flush against the surface, the pins **55** are initially at a lowest position on the face **51**. Then, the rotatable sleeve **57** is grasped and rotated, while holding the base **20** against the surface. During rotation of the sleeve **57**, which causes the face cam **50** to rotate, the pins **55** are driven along the face **51** until engaging in the grooves **53**. Meanwhile, the shaft **32**, driven to move up by the pins **55**, pulls only a central portion of the membrane **31** away from the contacting surface, as the edge of the membrane **31** is limited by the flange **45** of the shaft housing **41**, forming a vacuum seal between the membrane **31** and the contact surface. In this way, the suction base **20** is fixed on the surface.

The suction base **20** can be easily removed from the flat surface by rotating the rotatable sleeve **57**. In this way, the membrane **31** is driven to move down, the vacuum between the membrane **31** and the surface of the flat surface is eliminated, and the suction base **20** is no longer fixed on the flat surface.

Referring to FIGS. **6** and **7**, the load-supporting module **70** includes a housing **71** to house a pair of clamps **80**, a gear **91**, a torsion spring **92**, and a locking structure **95**.

The housing **71** is rectangle shaped, and includes a top cover **72** and a bottom cover **75**. The top cover **72** defines grooves **73** therein.

The clamps **80** are L-shaped, and includes vertical parts **81** and horizontal parts **82**. The horizontal parts **82** include rack gears **83**. The rack gears **83** define a plurality of locking grooves **84** on outer sidewalls thereof. The horizontal parts **82** are received by the grooves **73** of the top cover **72**. Referring to FIG. **1**, the vertical parts **81** extend upwardly from the housing **71**.

The gear **91** is rotatably mounted in the housing **71** and meshes with the rack gears **83** respectively. The torsion spring **92** fits over a shaft of the gear **91**. The torsion spring **92** causes the gear **91** to rotate, and causes the clamps **80** to mesh with the gear **91** to move away from each other.

The locking structure **95** is used to lock the clamps **80** and thus to fix the clamps **80** at a position, or to unlock the clamps **80** to allow the clamps **80** to move.

In the exemplary embodiment, the locking structure **95** includes a locking pin **96**, a spring **97**, and an elastic piece **98**. The locking pin **96** is slidably mounted in the housing **71**. The spring **97** fits over the locking pin **96** and causes the locking pin **96** to engage with a locking groove **84** of the clamp **80**, thus to lock the clamps **80**. The elastic piece **98** is rotatably mounted in the housing **71**. One end of the elastic piece **98** is fixed with the locking pin **96**. The other end of the elastic piece **98** stays in an opening **74** of the housing **71**. When the elastic piece **98** receives a pressure at the opening **74**, the elastic piece **98** causes the locking pin **96** to move away from the locking groove **84**, thus to unlock the clamps **80**.

Referring also to FIG. **1**, when loading, a device is placed on the housing **71**. The clamps **80** are pushed to move towards each other and clamp the device. The locking structure **95** locks the clamps **80**. In this way, the device is fixed on the load-supporting module **70**. When unloading, the elastic piece **98** of the locking structure **95** is pressed down and the clamps **80** are unlocked. The torsion spring **92** causes the

clamps **80** to move away from each other. Then the device can be unloaded from the load-supporting module **70**.

Moreover, it is to be understood that the invention may be embodied in other forms without departing from the spirit thereof. Thus, the present examples and embodiments are to be considered in all respects as illustrative and not restrictive, and the invention is not to be limited to the details given herein.

What is claimed is:

1. A device stand comprising:

a suction base comprising:

a suction piece comprising a membrane and a shaft extending upwardly from the membrane;

a fixing structure configured for fitting over the suction piece, comprising a shaft housing and a sleeve extending upwardly from the shaft housing, wherein the sleeve defines at least one axial running slot;

a face cam fitting over the sleeve of the fixing structure and defining a face thereon;

at least one pin, wherein one end of each pin engages with the face of the face cam, the other end of each of the pins passes through one of the at least one axial running slot of the sleeve and connects with the shaft of the suction piece; and

a fixing cap mounted on an upper end of the sleeve; and

a load-supporting module mounted on the suction base.

2. The device stand of claim **1**, wherein the face comprises depressed grooves on an upper end thereof for engaging with the at least one pin.

3. The device stand of claim **1**, further comprising a rotatable sleeve fitting over the face cam, configured for driving the face cam to rotate.

4. The device stand of claim **1**, wherein an upper end of the sleeve is closed.

5. The device stand of claim **4**, further comprising a spring disposed between the upper end of the sleeve and the shaft.

6. A suction base of a device stand, the suction base comprising:

a suction piece comprising a membrane and a shaft extending upwardly from the membrane;

a fixing structure configured for fitting over the suction piece, comprising a shaft housing and a sleeve extending upwardly from the shaft housing, wherein the sleeve defines a plurality of axial running slots;

a face cam fitting over the sleeve of the fixing structure and defining a face thereon;

a plurality of pins, wherein one end of each of the pins engages with the face of the face cam, the other end of each of the pins passes through one of the plurality of axial running slots of the sleeve and connects with the shaft of the suction piece; and

a fixing cap mounted on an upper end of the sleeve.

7. The suction base of claim **6**, wherein the face comprises depressed grooves on an upper end thereof for engaging with the plurality of pins.

8. The suction base of claim **6**, further comprising a rotatable sleeve fitting over the face cam, configured for driving the face cam to rotate.

9. The suction base of claim **6**, wherein an upper end of the sleeve is closed.

10. The suction base of claim **9**, further comprising a spring disposed between the upper end of the sleeve and the shaft.

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